

Forecasting Driving Exposure with Macroeconomic Factors: A PCA-Based Approach to Vehicle Miles Traveled

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Abstract

This paper examines whether macroeconomic information improves forecasts of vehicle miles traveled, which serves as a proxy for aggregate driving exposure in the context of auto insurance risk. Using monthly U.S. data from January 2000 to February 2026, five forecasting models were compared and analyzed: the naive mean model, the random walk model, the random walk with drift model, the PCA factor model, and the full macroeconomic regression model. After testing for stationarity, non-stationary variables were transformed using log differences, and principal component analysis was used to reduce the macroeconomic variables into fewer latent factors. Forecasts were then conducted using an expanding-window approach at one-month and twelve-month forecast horizons. The naive mean model produced the lowest overall forecasting errors; however, PCA proved to be the best alternative for predicting vehicle miles traveled. Since driving exposure is one component of auto insurance claim frequency, these findings suggest that forecasting VMT allows for discussions of claim risk and insurance pricing. Furthermore, the best forecast to inform said discussions is the simple naive mean model.

1. Introduction

The main variable of interest in this paper is vehicle miles traveled (VMT), which measures total driving activity in the economy. This variable is useful because it is closely related to auto insurance claim counts and driving exposure. In auto insurance, claim activity generally depends on how much people drive. More miles on the road mean more opportunities for accidents, which, in turn, increase the likelihood of claims. The intention was to measure direct insurance claim counts; however, this data is often proprietary and difficult to obtain publicly. VMT serves as a proxy for aggregate exposure to auto insurance risk. The goal is to forecast monthly VMT growth and evaluate whether macroeconomic information improves forecast accuracy. In other words, the idea is to use what is happening in the world and the broader economy to determine whether they are connected to driving behavior. For example, things like gas prices may affect how much people drive, and unemployment may affect commuting patterns. That being said, if these variables and others are linked to driving exposure, then, in theory, incorporating them into a model should improve forecasting accuracy.

In this paper, we will compare several different forecasting approaches. In forecasting experiments, the goal is to produce a large number of models and forecast each to determine which is most accurate. First, the best way to start is to establish simple models, such as a naive mean model, a random walk model, and a random walk with drift. These are also known as benchmark models and provide a standard for what constitutes a good forecast, because, in practice, simple time series models are often difficult to outperform. Next, this paper establishes a full macroeconomic regression model and estimates it using lagged macroeconomic and driving-related variables. Finally, we approach this forecast through a big data lens using principal component analysis to reduce the macroeconomic variables to a small number of latent factors. The main research question is whether principal component analysis improves forecasts of VMT growth relative to simple benchmark models and a regression model. The forecasts are evaluated using an expanding-window approach at one-month and twelve-month horizons. In other words, I am testing each model by predicting what will happen next month and next year, using only the information available at the time of the forecast, and then comparing it to what actually happened. Overall, establishing an accurate forecasting method for VMT enables analysis of changes in driving exposure, which is important for auto insurance risk assessment. While this does not directly measure claim count, a better understanding of driving exposure may help inform claims frequency, risk assessment, and insurance pricing.

2. Data

2.1 Data and Variables

The data used in this paper are monthly U.S. time series gathered from the Federal Reserve Economic Data (FRED) database. The dependent variable is vehicle miles traveled, which essentially serves as a proxy for driving exposure. The independent variables include general macroeconomic variables and driving-related variables: gas prices, auto sales, unemployment, the consumer price index, industrial production, and the federal funds rate. Altogether, these variables capture changes in driving costs, labor market conditions, inflation, and overall financial conditions. The sample selected from these variables spans January 2000 to February 2026, where most contained monthly observations. However, since gas prices were reported weekly, they were resampled to a monthly frequency by averaging the gas prices within each month. This dataset also contained some missing observations for each variable. Since the missing data were intermittent, they were imputed with the previous month's observed value. This part was imperative because the time series must be continuous for both forecasting and model comparison.

Variable	Description	Role
VMT	Vehicle Miles Traveled	Dependent variable
Gas price	Cost of driving	Macro/driving predictor
Auto sales	Vehicle demand	Macro/driving predictor
Unemployment	Labor market condition	Macro predictor
CPI	Inflation	Macro predictor
Industrial Production	Economic activity	Macro predictor
Federal Funds Rate	Financial conditions	Macro predictor

(Federal Reserve Bank of St. Louis, 2026)

2.2 Summary Statistics

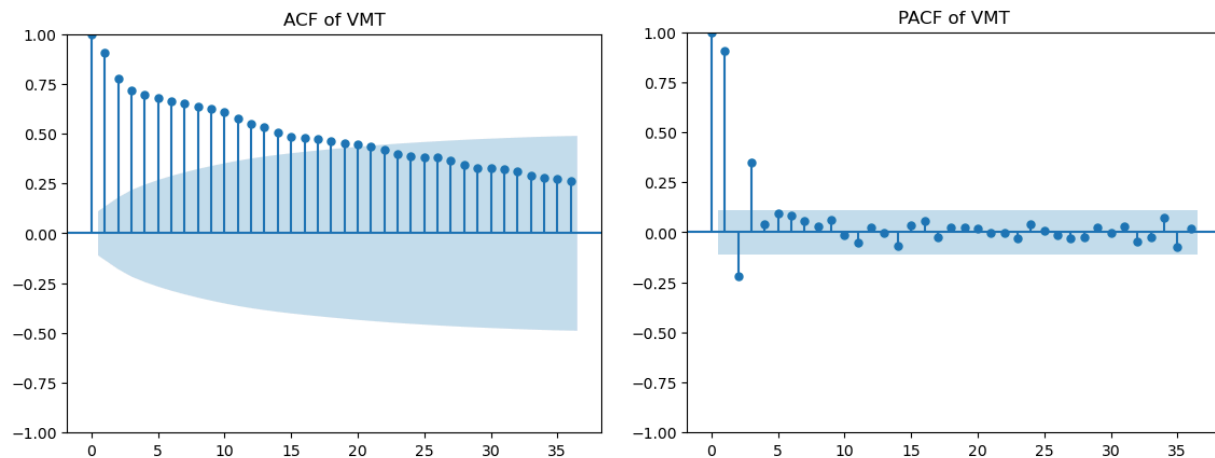
	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
VMT	314.0	254012.254777	14552.170562	167174.0000	245953.250000	250683.50000	267017.250000	285452.0000	-0.652894	3.161912
GASPRICE	314.0	2.656713	0.773258	1.0860	2.149125	2.71000	3.260312	4.9290	-0.022011	-0.722038
AUTOSALES	314.0	16.001061	2.162643	8.9190	15.189500	16.58500	17.484250	22.0550	-1.178619	1.258409
UNRATE	314.0	5.632166	1.939224	3.4000	4.200000	5.00000	6.200000	14.8000	1.316131	1.660781
CPI	314.0	234.062736	41.964892	169.3000	201.925000	231.73800	255.802000	327.4600	0.541866	-0.478378
INDPRO	314.0	97.263012	4.728157	84.5619	92.980950	98.94125	101.019875	104.1004	-0.728931	-0.569078
FEDFUNDS	314.0	2.018185	2.029514	0.0500	0.150000	1.26000	3.962500	6.5400	0.703952	-0.953430

The table above displays the summary statistics for the raw variables used in the analysis. A few important things to note are the differences in scale between the independent and dependent variables. For example, VMT has a mean of about 254,012, while gas prices only have a mean of 2.66. Therefore, standardizing these variables might be useful, especially given the principal component analysis that will follow this section. Furthermore, there was some interesting information contained in both the skewness and kurtosis due to the extreme values. For reference, skewness is a measure of the asymmetry of a probability distribution around its mean, and kurtosis is a measure of how often extreme outliers occur. The unemployment rate has a positive skewness of 1.32, which is quite high, and tends to exhibit extreme behavior during periods of economic stress. VMT also has relatively high kurtosis, indicating that this series exhibits intense fluctuations far from the mean. While this doesn't indicate any connection to other macroeconomic variables, it becomes much more interesting to explore because inference suggests that VMT doesn't respond randomly but is tied to current economic activity, and motivates a forecasting analysis.

2.3 Stationarity and Transformations

One important concept to address before estimating any forecasting model is stationarity, and whether the variables are stationary. The characteristics of a stationary time series are having a stable mean, variance, and autocorrelation structure over time. The main issue in regression models is spurious regression, which tends to occur with non-stationary variables. This occurs when two variables are completely unrelated, yet, when regressed on each other, they exhibit a meaningful relationship because they both move in the same direction (Granger & Newbold, 1974). For example, if VMT and another macroeconomic indicator both rise over the same sample period, a regression will show a relationship between them driven solely by that movement, rather than by true predictive information. The easiest fix for this problem is to convert non-stationary series into stationary ones, which means transforming the data. In this case, all non-stationary data were converted by taking the logarithm of the first difference, which, in simple terms, converts a raw series into monthly growth rates (Granger & Newbold, 1974). This yields a more stable mean and variance and essentially removes trend behavior, making it more suitable for forecasting.

Stationarity was examined using two methods: visually via autocorrelation and partial autocorrelation plots, and secondly via an Augmented Dickey-Fuller test (ADF). I deemed it necessary to examine at least one variable visually to properly explain the mechanics of a non-stationary series (Dickey & Fuller, 1979).



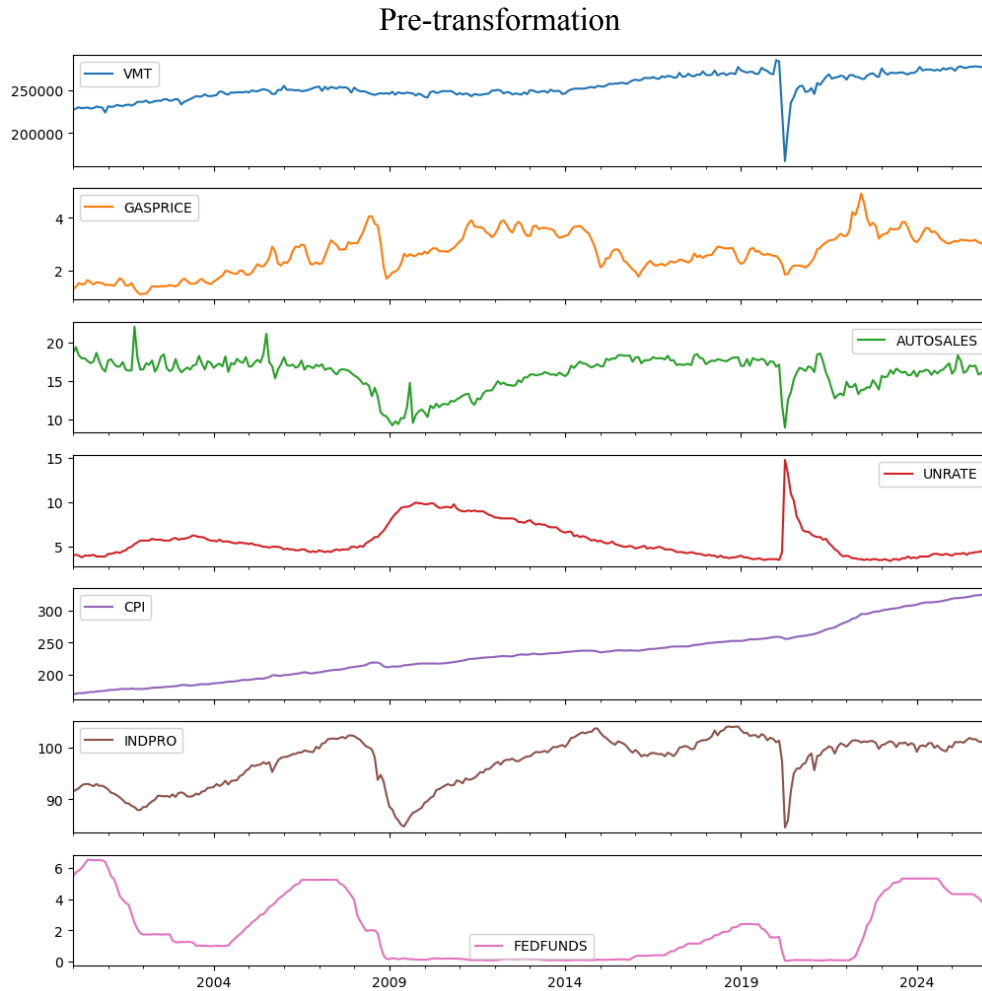
Displayed above are the autocorrelation and partial autocorrelation plots for VMT, and from these plots, it can be determined that this series is highly persistent. When a series is persistent, it means it depends on its past values and exhibits trendlike behavior. As stated before, this can prove to be problematic when building regression models; therefore, a transformation is necessary. Concerning the plots themselves, the autocorrelation plot shows a slow decline over many lags rather than dropping off quickly, which suggests that the current values of VMT are strongly related to the past values, a key indicator that there is a trend. Secondly, the partial autocorrelation plot shows a large spike at the first lag, indicating that the most recent

observation contains substantial information about the current value of the series, and that this pattern persists for the first few lags. After the first few, it falls within the confidence bands, which means that the dependence in this series is concentrated in the first few lags.

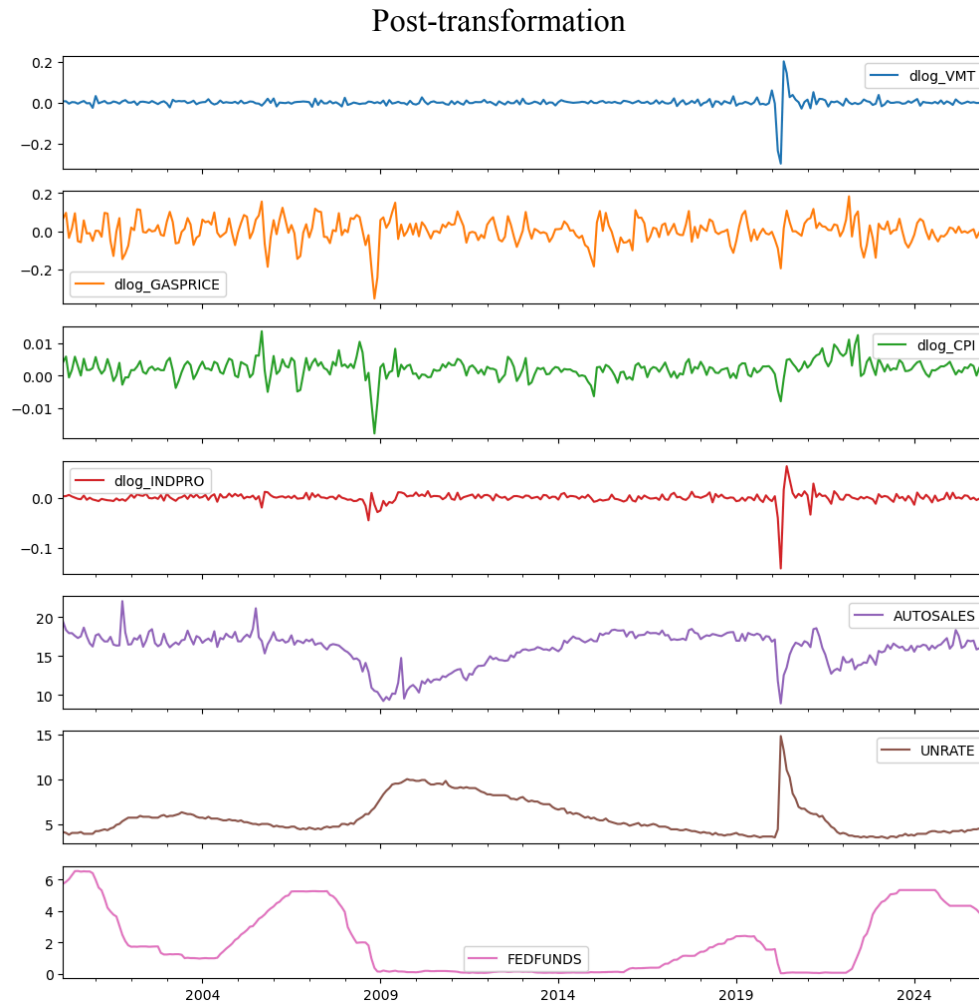
Secondly, an ADF test was run, which is another test for stationarity of the variables. The main statistic for determining whether a series is stationary in an ADF test is whether the p-value is above the 5% significance level (Dickey & Fuller, 1979). The table below shows the test results and suggests that VMT, gas prices, CPI, and industrial production are non-stationary in levels. These variables were then transformed using log differences. Subsequently, auto sales, unemployment, and the federal funds rate rejected the null at the 5% significance level, so the data remained as a level. Overall, this step was incredibly important because it reduces the risk of spurious regression and ensures that the models are comparing stationary relationships.

Variable	ADF p-value	Decision	Transformation
VMT	0.2661	Non-stationary	log-difference
Gas Price	0.2469	Non-stationary	log-difference
Auto Sales	0.0366	Stationary	level
Unemployment	0.0357	Stationary	level
CPI	0.9968	Non-stationary	log-difference
Industrial Production	0.2515	Non-stationary	log-difference
Federal Funds	0.0014	Stationary	level

2.4 Data Visualization



The above image depicts each time series variable before it underwent transformation. The raw data show that several variables display clearly trending behavior and large macroeconomic shocks. For example, VMT generally trends upward over the sample period but falls sharply during the 2020 COVID-19 period. The same could be said about many other variables in this time period.



The post-transformation plots look significantly different, as shown in the first four time series where the log difference was taken. Now those once-trending variables fluctuate more closely around a stable mean. However, the last three series remained unchanged because they were found to be stationary by the ADF tests and still exhibit economically meaningful movements. Overall, these plots support the decision to transform.

3. Methodology

3.1 Forecasting Framework

The framework for the following forecasts is to determine whether macroeconomic information improves forecasts of monthly VMT growth relative to simple benchmark forecasting models. The dependent variable in each forecasting model is $y_t = \Delta \log(\text{VMT}_t)$, representing the monthly growth rate of vehicle miles traveled. The analysis compares five forecasting approaches: a naive mean model, a random walk model, a random walk with drift model, a full macroeconomic regression model, and a PCA factor model. That said, the benchmark models serve as a reference point to determine whether the standard regression model and the PCA factor model add predictive value.

Furthermore, the forecasts are evaluated at two horizons: $h=1$, corresponding to predicting VMT one month from its current value, and $h=12$, corresponding to predicting VMT one year from its current value. Additionally, each model is estimated using an expanding-window approach. In essence, this means we are taking a sample from our current sample and applying the model to the available data. Then, after the first prediction, an expanding-window forecast adds the actual information back into the dataset until it reaches the present. Overall, each forecast uses as much information as possible to determine the predicted value, simulating an actual forecasting environment. Lastly, the forecast accuracy is then evaluated using the loss functions: mean squared forecast error (MSFE) and mean absolute error (MAE).

3.2 Benchmark Forecasting Models

In this forecasting exercise, I decided to include three simple benchmark models to serve as a baseline for comparing the more complex models. These models are a good benchmark because they are often hard to outperform in practice. In time-series forecasting, many economic series inherently exhibit persistence, and certain patterns aren't readily apparent when adding more predictors. Therefore, if the macroeconomic regression or PCA model cannot outperform these benchmarks, it does not provide meaningful forecasting insight. The first benchmark model is the naive mean model:

$$y_t = \mu + e_t$$

(Hyndman & Athanasopoulos, 2021)

Basically, this model says that future VMT growth is predicted by averaging past values. This model is best when the series that you are predicting fluctuates around a stable mean. The second benchmark model is the random walk model:

$$y_t = y_{t-1} + e_t$$

(Hyndman & Athanasopoulos, 2021)

This model is even simpler, since the forecast value equals the current value. This model is useful when predicting highly persistent series. Finally, the last model is the random walk with drift model:

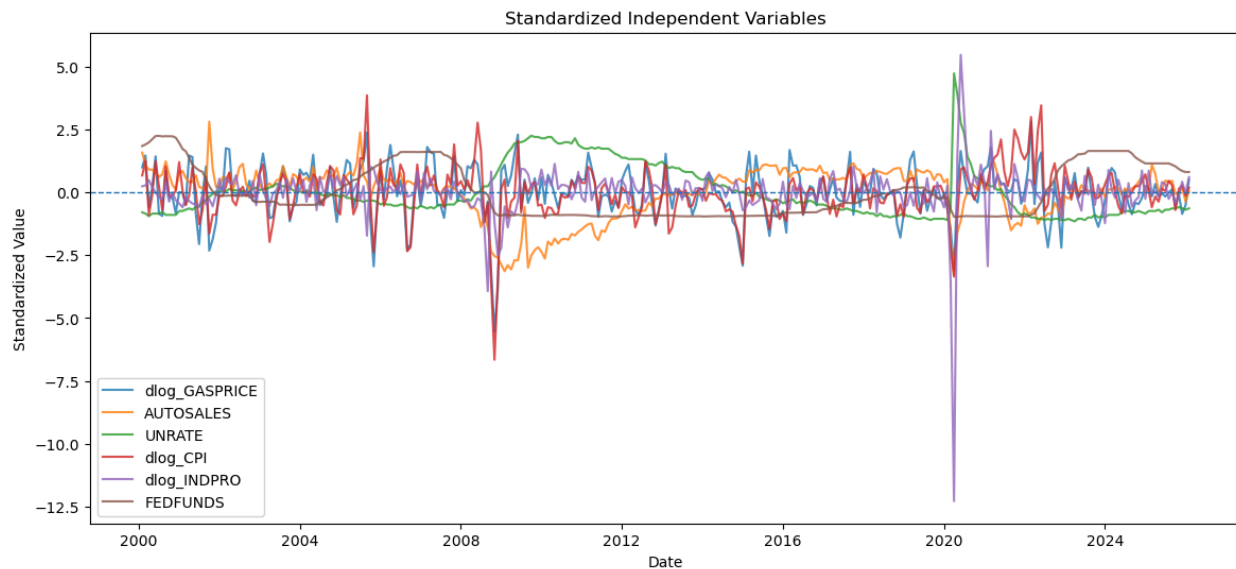
$$y_t = \mu + y_{t-1} + e_t$$

(Hyndman & Athanasopoulos, 2021)

This model is essentially the same as a random walk model, but with an extra term. The extra term is the average change in VMT growth across the entire sample and accounts for a persistent tendency in the series.

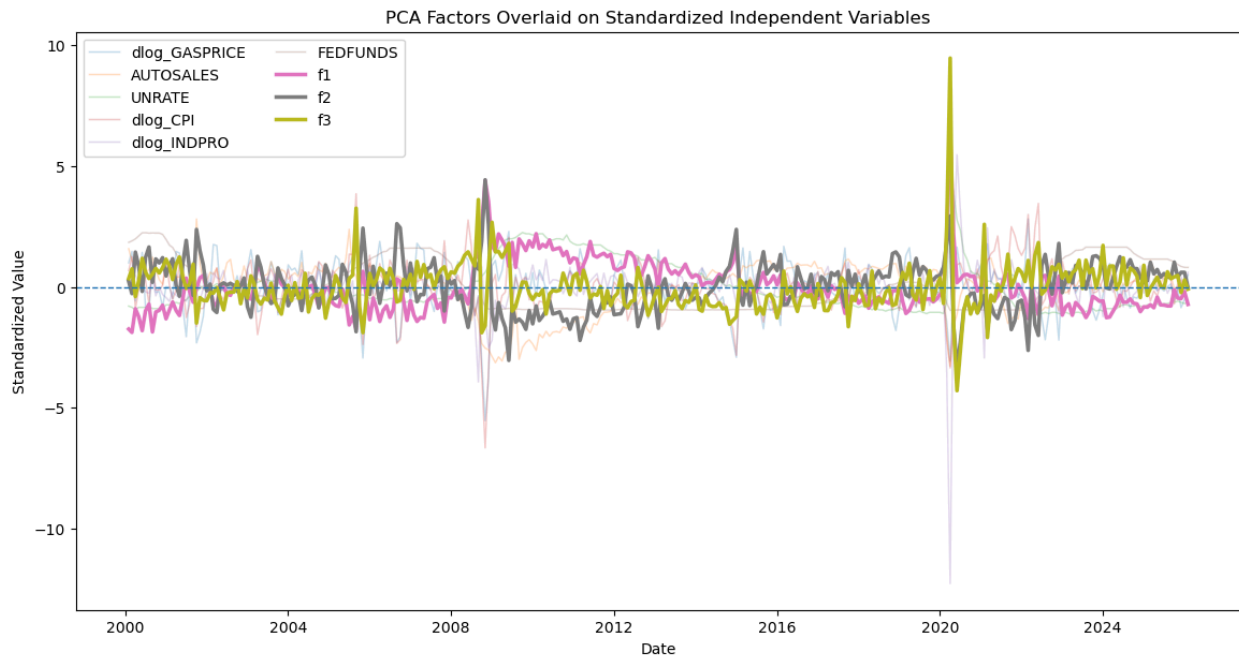
3.3 Principal Components Analysis and Factor Model

The idea behind principal components analysis (PCA) is that there are latent, unobserved variables that explain a large portion of the data's variation. Usually, this analysis and model are used for larger data problems, where you have thousands of predictors; however, in this case, they can still be useful. Essentially, this shrinks the model to only a couple of predictors, whereas including many may lead to variables containing the same information, which can introduce multicollinearity (Jolliffe, 2002). By using PCA, you reduce the original variables in the model to a smaller set of uncorrelated factors.



The process starts by first standardizing the independent variables so that each variable has a mean of zero and a standard deviation of one. This is an imperative step because PCA is very sensitive to differences in scale. The figure above represents the standardized independent variables. In the plot, there are several overlaps between the series as well as common

movements during the economic shock periods, such as the 2008 recession and the COVID-19 era. This proves that these variables contain common information that may be useful for predicting VMT growth. In simple terms, once you have gathered the standardized information, you construct a variance-covariance matrix and perform an eigenvalue decomposition, where you get eigenvalues associated with an eigenvector. The eigenvectors determine the weights used to build each principal component, and the eigenvalues tell us how much variation each component explains (Jolliffe, 2002). All this really means is that a couple of these results serve as a summary variable that captures the main patterns of the data.



The image above displays the estimated PCA factors overlaid on the standardized independent variables. As you can see, these factors compress the macroeconomic information into a smaller number of predictors that seem to explain most of the data's variance. The full model is defined as follows:

$$\Delta \log(\text{VMT}_t) = \alpha + \beta_1(\Delta \log(\text{VMT}_{t-1})) + \gamma_1 f_{1,t-1} + \gamma_2 f_{2,t-1} + \gamma_3 f_{3,t-1} + \sum_{m=2}^{12} \delta_m D_{m,t} + e_t$$

In this model, y_{t-1} represents the lagged VMT growth, f_1 , f_2 , and f_3 are the lagged PCA factors, and $D_{m,t}$ are the monthly dummy variables. The reason I included monthly dummy variables in this model is to control for seasonality, since the data is monthly, and driving patterns may be seasonal. For example, driving behavior might change during holidays or summer travel. This is imperative because, if there is seasonality, the PCA factors aren't forced to account for it.

Furthermore, I included the lag for each term to ensure the model uses only information available at the time of the forecast. Finally, we get the regression output:

OLS Regression Results						
=====						
Dep. Variable:	dlog_VMT	R-squared:	0.159			
Model:	OLS	Adj. R-squared:	0.116			
Method:	Least Squares	F-statistic:	3.728			
Date:	Sun, 03 May 2026	Prob (F-statistic):	5.00e-06			
Time:	15:31:25	Log-Likelihood:	704.95			
No. Observations:	312	AIC:	-1378.			
Df Residuals:	296	BIC:	-1318.			
Df Model:	15					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	0.0068	0.005	1.311	0.191	-0.003	0.017
dlog_VMT_lag1	0.4275	0.072	5.897	0.000	0.285	0.570
f1_lag1	0.0043	0.001	4.114	0.000	0.002	0.006
f2_lag1	0.0014	0.001	1.147	0.252	-0.001	0.004
f3_lag1	0.0104	0.002	5.296	0.000	0.007	0.014
month_2	-0.0140	0.007	-1.922	0.056	-0.028	0.000
month_3	-0.0107	0.007	-1.459	0.146	-0.025	0.004
month_4	-0.0136	0.007	-1.839	0.067	-0.028	0.001
month_5	-0.0006	0.007	-0.082	0.935	-0.015	0.014
month_6	-0.0031	0.007	-0.427	0.669	-0.017	0.011
month_7	-0.0053	0.007	-0.727	0.468	-0.020	0.009
month_8	-0.0029	0.007	-0.396	0.693	-0.017	0.011
month_9	-0.0070	0.007	-0.966	0.335	-0.021	0.007
month_10	-0.0078	0.007	-1.072	0.285	-0.022	0.007
month_11	-0.0070	0.007	-0.968	0.334	-0.021	0.007
month_12	-0.0050	0.007	-0.698	0.486	-0.019	0.009
=====						
Omnibus:	294.583	Durbin-Watson:	1.778			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	29193.870			
Skew:	-3.335	Prob(JB):	0.00			
Kurtosis:	49.917	Cond. No.	73.6			
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Ultimately, we get quite unpromising regression results with a low R^2 and little statistical significance in our coefficients; however, this is not about model accuracy, so these results do not matter. All that we care about is forecasting accuracy, which we will measure shortly.

3.4 Full Macroeconomic Regression Model

OLS Regression Results						
=====						
Dep. Variable:	dlog_VMT	R-squared:	0.278			
Model:	OLS	Adj. R-squared:	0.233			
Method:	Least Squares	F-statistic:	6.261			
Date:	Sun, 03 May 2026	Prob (F-statistic):	5.43e-13			
Time:	15:31:25	Log-Likelihood:	728.73			
No. Observations:	312	AIC:	-1419.			
Df Residuals:	293	BIC:	-1348.			
Df Model:	18					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.0999	0.021	-4.679	0.000	-0.142	-0.058
dlog_VMT_lag1	0.4746	0.071	6.703	0.000	0.335	0.614
dlog_GASPRICE_lag1	-0.0717	0.049	-1.452	0.148	-0.169	0.026
AUTOSALES_lag1	0.0034	0.001	3.836	0.000	0.002	0.005
UNRATE_lag1	0.0076	0.001	6.318	0.000	0.005	0.010
dlog_CPI_lag1	1.8985	0.942	2.015	0.045	0.044	3.753
dlog_INDPRO_lag1	-1.2385	0.174	-7.099	0.000	-1.582	-0.895
FEDFUNDS_lag1	0.0025	0.001	2.882	0.004	0.001	0.004
month_2	-0.0123	0.007	-1.759	0.080	-0.026	0.001
month_3	-0.0081	0.007	-1.135	0.257	-0.022	0.006
month_4	-0.0078	0.008	-0.980	0.328	-0.023	0.008
month_5	0.0008	0.008	0.101	0.920	-0.014	0.016
month_6	-0.0008	0.007	-0.103	0.918	-0.015	0.014
month_7	-0.0048	0.007	-0.699	0.485	-0.018	0.009
month_8	-0.0039	0.007	-0.577	0.564	-0.017	0.009
month_9	-0.0081	0.007	-1.192	0.234	-0.021	0.005
month_10	-0.0087	0.007	-1.267	0.206	-0.022	0.005
month_11	-0.0063	0.007	-0.937	0.349	-0.020	0.007
month_12	-0.0039	0.007	-0.575	0.565	-0.017	0.009
=====						
Omnibus:	305.664	Durbin-Watson:	1.763			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	20377.982			
Skew:	-3.726	Prob(JB):	0.00			
Kurtosis:	41.885	Cond. No.	1.19e+04			
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The following regression results show a little bit more promise than its factor model counterpart. The R^2 has improved, and more coefficients are statistically significant. However, this doesn't really matter, and these insights don't require much analysis, since forecasting accuracy is the key measure, not the model's goodness of fit.

3.5 Forecasting Evaluation

Finally, after estimating the benchmark models, the macroeconomic regression, and the PCA factor model, each model is evaluated based on out-of-sample forecast accuracy. Again, this is the most imperative part because it doesn't really matter if the model doesn't fit the historical data well, as long as it leads to more accurate forecasts, it is deemed useful. Because the forecasts were generated using an expanding-window approach, I defined the initial training sample from January 2000 to December 2018. Subsequently, each model produced a forecast either 1 month or 12 months ahead. After a forecast was generated, the training sample was expanded by 1 month, and the process was repeated.

Each forecast was assessed using the MSFE and MAE loss functions. Intuitively, each is a function of the error term, defined as the difference between the forecasted and actual values.

$$e_{t+h} = \hat{y}_{t+h|t} - y_{t+h}$$

$$MSFE = \frac{1}{T} \sum_{t=1}^T e_{t+h}^2$$

$$MAE = \frac{1}{T} \sum_{t=1}^T |e_{t+h}|$$

Both metrics measure forecast accuracy, but they emphasize the forecasting errors differently. Since MSFE squares the forecasting errors, it penalizes bigger mistakes more heavily. On the other hand, MAE measures the average absolute size of the forecast error. The key is that these errors are all positive, preventing positive and negative forecasting errors from canceling each other out. Overall, the metric with the lowest value indicates the best-performing forecasting model.

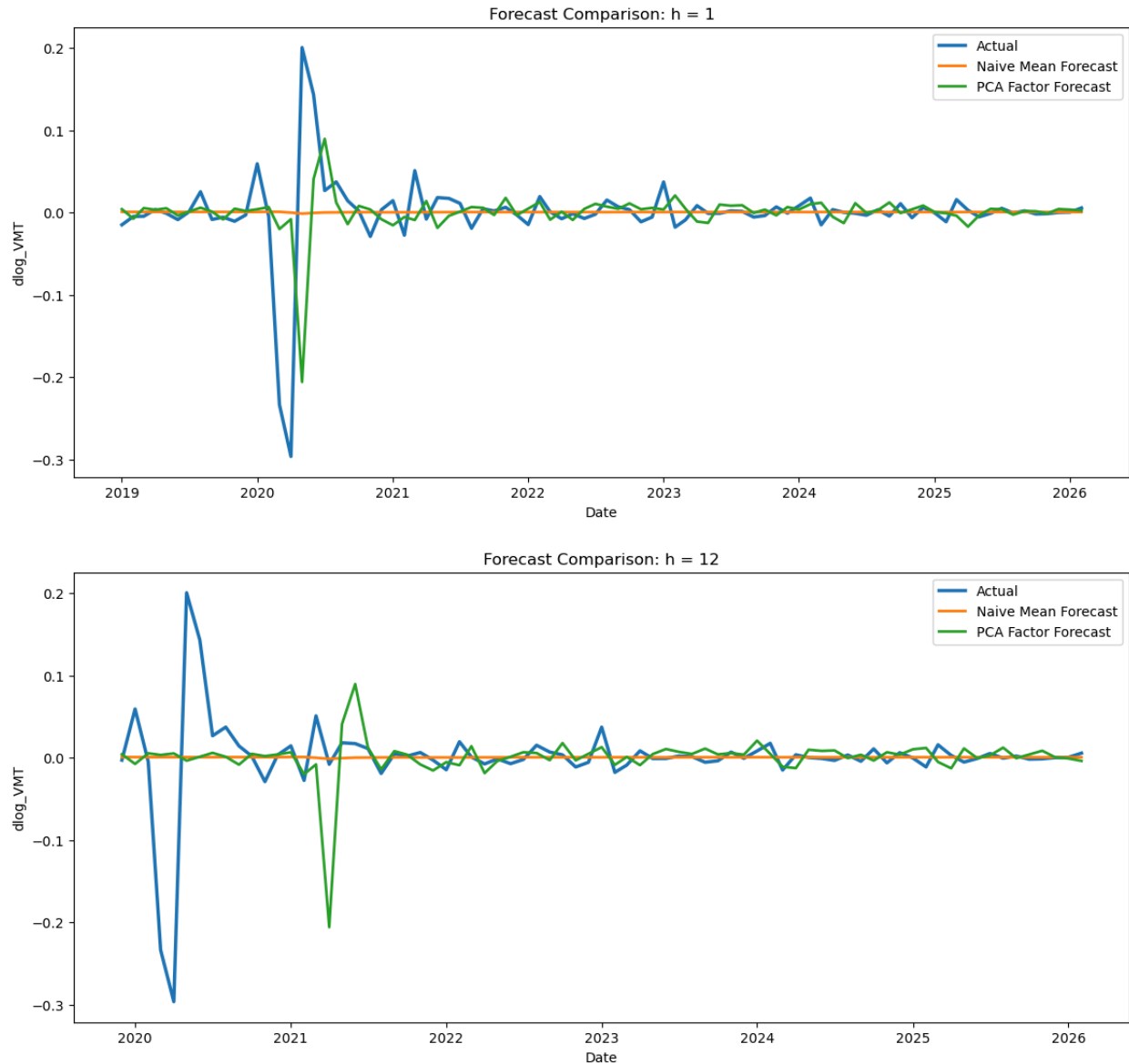
4. Results

4.1 Forecast Accuracy

Model	h=1 MSFE	h=1 MAE	h=12 MSFE	h=12 MAE
Naive Mean	0.002574	0.019261	0.002924	0.020793
Random Walk	0.004194	0.025421	0.006052	0.037788
Random Walk w/ Drift	0.004211	0.025470	0.006360	0.038757
PCA Factor Model	0.003857	0.024069	0.003637	0.026519
Full Macro Regression	0.004008	0.027715	0.004100	0.033341

The table above shows the forecast accuracy results for each model at the one-month-ahead and twelve-month-ahead forecast horizons. At the one-month-ahead horizon, the naive mean model performs the best with an MSFE of 0.002574 and an MAE of 0.019261. This implies that monthly VMT growth is hard to improve upon using more complex models. The random walk and random walk with drift models performed significantly worse than all other models. This is likely because the data were transformed into a stationary series, removing much of the persistence. Among the more complex models, the PCA factor model performs better than the full macroeconomic model at the one-month horizon. This suggests that reducing macroeconomic variables to a small number of latent factors yields a cleaner forecast specification. The same could be said about the twelve-month-ahead forecasts as well. Overall, the results show that the naive mean model is the strongest performing forecasting model; however, the PCA factor model consistently outperformed both the full macroeconomic regression model and the random walk models. Furthermore, the difference between the naive model and factor model loss function results was not significantly different. Although including factors does not outperform the simple model, the PCA model could still be useful for summarizing economic conditions, improving on more direct macroeconomic regression approaches.

4.2 Forecast Visualization



The two plots provide a visual comparison between the two best-performing models, the naive mean model and the PCA factor model, against the actual values of monthly VMT growth with the forecasts produced. If we look at the $h=1$ forecasts, the actual series is fairly stable for most of the series, except for the extreme shock in 2020 during the COVID-19 period. The naive mean forecast is almost flat throughout the evaluation period and does not capture short-run fluctuations. The PCA factor forecasts are more responsive to changes, yet they still do not outperform the naive mean model. In the $h=12$ forecasts, the same general pattern appears; however, the difference between the actual series and the factor model is more noticeable. This is expected because at the twelve-month horizon, forecasts become more difficult.

4.3 Robustness Check

Another necessary step in evaluating forecasting models is to conduct robustness checks to assess whether the main results are sensitive to changes in the modeling setup. In this case, I tested the model performance by changing the in-sample period from January 2000 to December 2018 to January 2000 to December 2015. If the same pattern holds across different sample periods, the findings are more reliable. The table below reports the one-month-ahead and twelve-month-ahead forecast horizons and their loss function results.

Model	h=1 MSFE	h=1 MAE	h=12 MSFE	h=12 MAE
Naive Mean	0.001845	0.015809	0.002018	0.016876
Random Walk	0.003036	0.021775	0.004123	0.027935
Random Walk w/ Drift	0.003049	0.021820	0.004332	0.028593
PCA Factor Model	0.002752	0.019276	0.002500	0.020788
Full Macro Regression	0.002858	0.021857	0.002812	0.025350

Overall, the model results remain consistent with the original findings, meaning the robustness check verifies that the naive mean forecast is the strongest forecasting model. Furthermore, it also reinforces the idea that the PCA model outperforms all subsequent models and is the best complex model for forecasting VMT.

5. Discussion and Conclusion

The results of this paper show that the naive mean model produced the strongest forecast for monthly VMT growth at both the one-month-ahead and twelve-month-ahead forecast horizons. This implies that the monthly changes are difficult to forecast using recent values or macroeconomic predictors. This is likely because VMT growth tends to fluctuate around a stable average for much of the sample. Although the naive mean model performed the best overall, the PCA factor model produced the strongest results among the models that incorporated additional macroeconomic information. At both forecasting horizons, the PCA factor model outperformed every other model. Again, even though the simple benchmark model was superior,

macroeconomic variables, when summarized into latent factors, may contain useful information about driving exposure.

These findings are relevant to the auto insurance context because VMT represents one important component of claim frequency risk. More miles driven generally create more opportunities for accidents and claims. While VMT does not directly measure claim count, it captures aggregate driving exposure, which is closely related to the environment in which claims arise. Therefore, forecasting VMT could be useful for discussing driving exposure trends and insurance pricing. However, the results indicate that simple historical patterns are the most informative measure for forecasting, and adding macroeconomic information does not automatically improve forecast accuracy. There are several limitations to this analysis. The most important limitation is that VMT is only a proxy for actual auto insurance claim activity. Unfortunately, most data sources are proprietary or lack a time-series component. In addition, the COVID-19 period created an unusually large shock to driving activity, which probably hurt forecasting accuracy for some models. In the future, I would like to add additional robustness checks in each model that control for the COVID-19 shock.

6. References

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